Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/02/2013
PI Name:	Porterfield, D. Marshall Ph.D.		
Project Title:	Sporesat Mission: Investigating biophysical mechanisms of plant gravisensing using a lab-on-a-chip approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture (2) Bioregenerative Life Support		
PI Email:	porterf@purdue.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(765) 494-1190
Organization Name:	Purdue University		
PI Address 1:	302 W. Wood Street, Young Hall		
PI Address 2:			
PI Web Page:			
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2108	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	03/01/2011	**End Date:**	08/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Zell, Peter	**Contact Phone:**	(650) 604-3690
Contact Email:	peter.t.zell@nasa.gov		
Flight Program:	Small Satellites		
Flight Assignment:			
Key Personnel Changes/Previous PI:	PI change from Marshall Porterfield to Jenna Rickus in November 2012. Amani Salim replaced Porterfield initially until November 2012 and she remains as Co-PI on the new project with Rickus as the PI.		
COI Name (Institution):	Roux, Stanley (University of Texas, Austin)		
Grant/Contract No.:	NNX11AF48A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A detailed understanding of the molecular and biophysical mechanisms for plant gravity sensing still eludes the scientific community. To gain deeper knowledge of the mechanism of cell gravity sensing, we will study activation of cell gravisensing and electrophysiological signaling in a single-cell model system (Ceratopteris richardii spores) using a "lab-on-a-chip" microsensor technology platform. We here propose a Small Complete Mission (SCM) based on a Mission of Opportunity-2 nanosatellite we call SporeSat, which will adapt hardware and technologies from previous and current missions including: GeneSat, PharmaSat, O/OREOS and GraviSat. Our approach allows for real-time measurement of calcium signaling and maintenance of variable gravity treatments during the flight experiment. Our science objectives further current work that has documented molecular components of biomechanical transduction. Our technical objectives focus on refining our current lab-on-a-chip sensor system to build a new CEL-C Advanced bioCD designed for the micro-satellite platform. This project contributes to NASA's initiatives in the areas of both Astrobiology and Fundamental Space Biology and will advance the science and technology capabilities for future research and development of astronaut health programs as well as those for advanced life support for exploration in the space environment, the moon, and on other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	PI change to Jenna Rickus in November 2012. Previous PI was Marshall Porterfield; Amani Salim replaced him initially until November 2012 and she remains as Co-PI on the new project with Rickus as the PI.. End date for task with PI=Porterfield changed to 8/31/2012 (from 2/28/2014)--See Rickus (PI) for future reports; project continues until 2/28/2014 (Ed., as of 12/2012).
Bibliography Type:	Description: (Last Updated: 03/29/2017)
Articles in Peer-reviewed Journals	Park J, Salmi ML, Wan Salim WW, Rademacher A, Wickizer B, Schooley A, Benton J, Cantero A, Argote PF, Ren M, Zhang M, Porterfield DM, Ricco AJ, Roux SJ, Rickus JL. "An autonomous lab on a chip for space flight calibration of gravity-induced transcellular calcium polarization in single-cell fern spores." Lab Chip. 2017 Mar 14;17(6):1095-103. http://dx.doi.org/10.1039/c6lc01370h ; PubMed PMID: 28205656 , Mar-2017